what are the gaps existing

Research Gaps in PURE and Enterprise Upgrading

The literature on Productive Use of Renewable Energy (PURE) and enterprise upgrading is fragmented across two largely disconnected research streams, leaving several critical gaps at their intersection (Aarakit et al., 2024; Lema et al., 2025).

Evidence Coverage Across PURE and Upgrading

	Developing country field studies	Macro-econometric analysis	Policy & institutional analysis	Gender & equity lens
PURE income & livelihoods	GAP	GAP	GAP	GAP
Firm-level productivity	GAP	GAP	GAP	GAP
Industrial structure upgrading	GAP	GAP	GAP	GAP
GVC participation & upgrading	GAP	GAP	GAP	GAP
PURE financing mechanisms	GAP	GAP	GAP	GAP

The most striking gap is the near-total absence of studies linking PURE adoption at the micro-enterprise level to firm-level upgrading outcomes such as productivity growth, product complexity, or value-chain position. PURE studies measure income and yield changes for smallholders and micro-enterprises (Shastry & Rai, 2025; Terrapon-Pfaff et al., 2017), while firm-upgrading research operates at the sector or listed-company level using patent counts, total factor productivity, or industrial structure indices (Su & Fan, 2022; Lin & Zhu, 2024; Wen et al., 2021). No retrieved paper bridges these scales directly.

Definitional and Conceptual Gaps

The literature lacks a consensus definition of "productive use of energy," with scholars and policymakers employing the term across different energy carriers, end uses, and activity types (Aarakit et al., 2024). Most studies focus on productive use of electricity for income generation, though some adopt a broader perspective encompassing electric and non-electric energy (Aarakit et al., 2024). Very few renewable energy projects explicitly target productive end-use as an objective, focusing instead on rural electrification for consumption uses such as lighting and phone charging (Terrapon-Pfaff et al., 2017). This definitional ambiguity means that evidence linking PURE specifically to enterprise upgrading—rather than to general income generation—remains thin (Aarakit et al., 2024).

Methodological and Scale Gaps

- **Micro-macro disconnect**: PURE field studies use qualitative methods with small samples of rural entrepreneurs (Shastry & Rai, 2025), while upgrading studies use panel data of listed firms across provinces or countries (Su & Fan, 2022; Qamri et al., 2025), with no study tracing PURE adoption through to measurable upgrading outcomes.
- **Geographic concentration**: Firm-level upgrading evidence is dominated by China (Su & Fan, 2022; Lin & Zhu, 2024; Wen et al., 2021), while PURE field studies cluster in Sub-Saharan Africa and South Asia (Ngowa et al., 2024; Bhanjdeo et al., 2026; Pueyo et al., 2020), limiting cross-regional generalization.
- **Temporal shortfalls**: Most PURE impact evaluations capture short-term outcomes within two years of intervention (Terrapon-Pfaff et al., 2017), whereas industrial upgrading unfolds over longer horizons that no PURE study tracks (Lu et al., 2026).
- **Gender and equity underexamination**: Only one study explicitly examines how gender shapes productive energy use and enterprise outcomes, finding women entrepreneurs benefit less from electricity-based productive use than men (Pueyo et al., 2020).

Outcome Measurement Gaps

PURE studies predominantly measure income, yield, or livelihood resilience (Shastry & Rai, 2025), rarely tracking whether enterprises upgrade their production processes, move into higher-value-added activities, or shift position within value chains. The GVC upgrading literature, by contrast, uses metrics like technological complexity, product space density, and value addition (Lu et al., 2026; Lema et al., 2025), but does not examine whether renewable energy adoption drives these transitions at the enterprise level. Green credit and fiscal policy studies measure R&D intensity and total factor productivity but focus on energy-intensive and listed firms rather than the small and micro-enterprises that dominate PURE adoption (Wen et al., 2021; Zhao et al., 2021). Supply chain configuration research shows diversification improves renewable energy firm TFP through asset turnover and knowledge absorption (Lin & Zhu, 2024), yet no study extends this logic to PURE-adopting micro-enterprises.

Causal Mechanism and Contextual Gaps

Gap Area	What Is Missing	Closest Evidence
PURE-to-upgrading causal chain	No study traces renewable energy adoption through productivity gains to value-chain upgrading	(Terrapon-Pfaff et al., 2017; Lema et al., 2025)
Complementary inputs	Energy is necessary but insufficient; bundling with training, finance, and market access is noted but not rigorously tested for upgrading outcomes	(Terrapon-Pfaff et al., 2017; Aarakit et al., 2024)
Negative or null effects	Mini-grid studies show no business improvement, but no study examines whether failed PURE adoption retards upgrading	(Terrapon-Pfaff et al., 2017)
Community enterprise scaling	Institutional complexity in scaling is conceptualized but not linked to productive upgrading outcomes	(Bauwens et al., 2022)

Gap Area	What Is Missing	Closest Evidence
Post-COVID and geopolitical shifts	Panel data predates supply chain realignments; no PURE study accounts for recent global energy landscape uncertainty	(Lu et al., 2026)

FIGURE 1 Key research gaps at the PURE–enterprise upgrading intersection and their closest available evidence

The field would benefit from longitudinal studies that track PURE-adopting enterprises over multiple years using upgrading-specific metrics—product complexity, value addition, and value-chain position—rather than stopping at income or yield measures, and that connect micro-level PURE outcomes to the macro-level industrial upgrading frameworks already established in the GVC and innovation policy literatures (Lema et al., 2025; Lu et al., 2026).

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